AI-POWERED CUSTOMER CHURN PREDICTION AND RETENTION SYSTEM

A Project Based Learning Report Submitted in partial fulfilment of the requirements for the award of the degree

of

Bachelor of Technology

in The Department of Artificial Intelligence and Data Science

Machine Learning 25SC2107E

Submitted by

2510080029: Oddesar Sudeeptha

2510080037: Lokrey Latika

2510080043: Samineni Sai Siri

2510080061:Muvva Kovidha

Under the guidance of

Dr. Gangamohan Paidi



Department of Electronics and Communication Engineering

Koneru Lakshmaiah Education Foundation, Aziz Nagar

Aziz Nagar – 500075

July - 2026.

ABSTRACT

Customer churn is a major challenge for organizations that provide subscription-based or long-term services. When customers discontinue a service, organizations may lose recurring revenue and may need to spend additional resources to acquire new customers. Identifying customers who are likely to leave before they actually churn can help organizations take appropriate retention measures.

This project presents an AI-Powered Customer Churn Prediction and Retention System that uses machine learning to predict whether a customer is likely to discontinue a service. The project uses the Telco Customer Churn dataset, which contains customer demographic information, service details, contract information, billing information, tenure, and churn status.

The proposed system follows a machine learning pipeline consisting of data collection, preprocessing, exploratory data analysis, feature preparation, model training, evaluation, and inference. Three machine learning algorithms, namely Logistic Regression, Decision Tree, and Random Forest, were trained and evaluated using accuracy, precision, recall, and F1-score. The models were compared, and the best-performing model was selected based on its F1-score for customer churn prediction.

The selected model is integrated with a Gradio-based graphical user interface that allows users to enter customer details and obtain a churn prediction along with the estimated churn probability. The system demonstrates how machine learning can help identify potentially high-risk customers and support customer retention decisions.

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AI-POWERED CUSTOMER CHURN PREDICTION AND RETENTION SYSTEM

CHAPTER 1: INTRODUCTION

Customer churn is an important problem for organizations that provide subscription-based services. When customers stop using a service, organizations may lose regular revenue and spend additional resources to acquire new customers. Therefore, identifying customers who are likely to leave can help organizations take suitable actions to improve customer retention.

Traditional methods of identifying customer churn often depend on manually analyzing customer information. However, when the number of customers becomes large, it becomes difficult to identify important patterns manually. **Machine learning** can help solve this problem by learning patterns from historical customer data and using them to predict the behavior of new customers.

This project presents an **AI-Powered Customer Churn Prediction and Retention System** using machine learning. The system uses the **Telco Customer Churn dataset**, which contains information about customer demographics, services, contracts, tenure, billing, payment methods, and churn status.

The project uses supervised machine learning algorithms to classify customers into two categories: **likely to churn** and **likely to stay**. Logistic Regression, Decision Tree, and Random Forest models are trained and compared to determine the most suitable model for the prediction task.

The final system also performs **inference** on customer information and provides a churn probability. A Gradio-based graphical interface is developed so that users can enter customer details and receive predictions without directly using the Python code.

1.1 Problem Statement

Organizations need an automated method to identify customers who are at risk of leaving their service. The problem addressed in this project is to develop a machine learning classification system capable of predicting customer churn from historical customer and service-related information.

1.2 Objectives

The main objectives of the project are:

1. To study customer information and identify features relevant to churn prediction.
2. To preprocess the customer churn dataset for machine learning.
3. To perform exploratory data analysis to understand the dataset.
4. To train multiple machine learning classification algorithms.
5. To compare Logistic Regression, Decision Tree, and Random Forest.
6. To evaluate the trained models using appropriate performance metrics.
7. To select a suitable model for customer churn prediction.
8. To perform inference on previously unseen customer records.
9. To develop a simple graphical interface for customer churn prediction.
10. To provide churn probability along with the predicted class.

1.3 Scope

The scope of this project is to develop a machine learning system for predicting customer churn using customer and service-related data. It helps identify high-risk customers and supports better retention decisions. The system includes data preprocessing, model training, evaluation, inference, and a simple GUI.

1.4 Proposed Solution

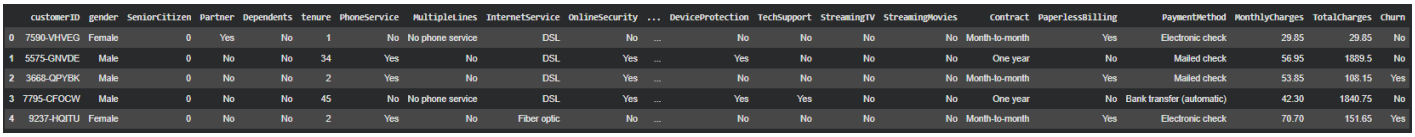
The proposed solution uses machine learning models such as Logistic Regression, Decision Tree, and Random Forest to predict customer churn. The best-performing model is selected for prediction. A Gradio-based GUI allows users to enter customer details and obtain the churn prediction and probability.

CHAPTER 2: METHODOLOGY

The methodology describes the sequence of operations used to develop the customer churn prediction system. The project follows a supervised machine learning approach because the dataset contains a known target variable representing whether a customer churned.

2.1 Dataset Description

The project uses the Telco Customer Churn dataset downloaded using KaggleHub. The dataset contains customer-level information related to demographic characteristics, services, contracts, billing, payment methods, tenure, and churn.



	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No	One year	No	Mailed check	56.95	1889.5	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year	No	Bank transfer (automatic)	42.30	1840.75	No
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes

Figure 1: Dataset Description

Important features include:

- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection

- Tech Support
- Streaming TV
- Streaming Movies
- Contract
- Paperless Billing
- Payment Method
- Monthly Charges
- Total Charges
- Churn

The Churn attribute is the target variable.

2.2 Data Preprocessing

Data preprocessing was performed before model training. The customerID column was removed because it is a unique identifier. The TotalCharges column was converted to numeric form, and missing values were handled using median imputation. Categorical features were converted into numerical representations using One-Hot Encoding, while numerical features were standardized using StandardScaler. A preprocessing pipeline was used to apply these transformations consistently during training and prediction.

	SeniorCitizen	Partner	Dependents	tenure	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	0	1	0	1	1	0	0	2	0	0	0	0	0	1	2	29.85	29.85	0
1	0	0	0	34	0	0	2	0	2	0	0	0	1	0	3	56.95	1889.50	0
2	0	0	0	2	0	0	2	2	0	0	0	0	0	1	3	53.85	108.15	1
3	0	0	0	45	1	0	2	0	2	2	0	0	1	0	0	42.30	1840.75	0
4	0	0	0	2	0	1	0	0	0	0	0	0	0	1	2	70.70	151.65	1

Figure 2: Data Preprocessing

2.3 Exploratory Data Analysis

EDA was performed to understand the dataset and identify important patterns. The analysis included dataset dimensions, data types, missing values, descriptive statistics, churn distribution, and feature relationships.

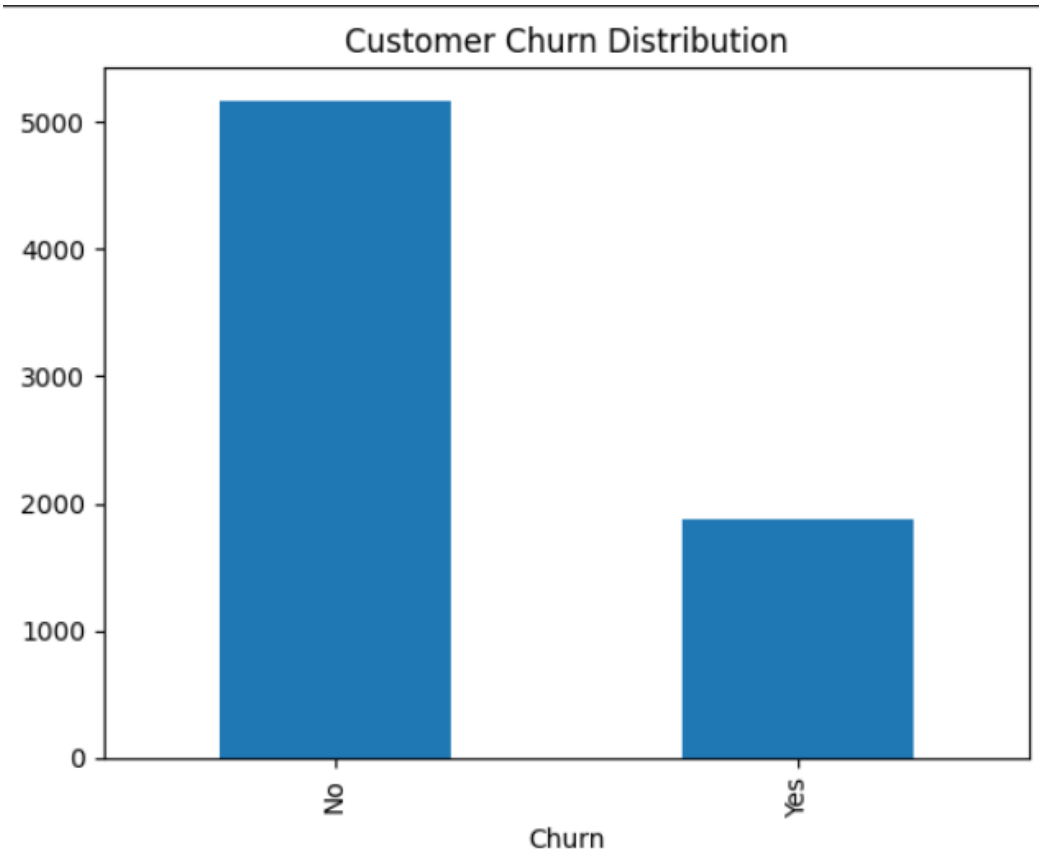


Figure 3: Churn Distribution Visualization.

2.4 Train-Test Split

The processed dataset was divided into training and testing subsets.

Training Data: 80%

Testing Data: 20%

The training data was used to learn patterns, while the test data was used to evaluate how the trained model performs on unseen records.

2.5 Machine Learning Algorithms

Logistic Regression

Logistic Regression was used as a binary classification model. It estimates the probability that a customer belongs to the churn class.

The model provides both:

- Predicted class
- Churn probability

Decision Tree

Decision Tree classification creates a tree-like structure of decisions based on feature values. It can model nonlinear relationships between customer attributes and churn.

Random Forest

Random Forest combines multiple decision trees to produce a stronger classification model. Each tree contributes to the final prediction.

2.6 Model Training

Each model was trained using the training dataset. After training, predictions were generated using the test dataset.

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

2.7 Inference

Inference is the process of applying the trained model to new customer data.

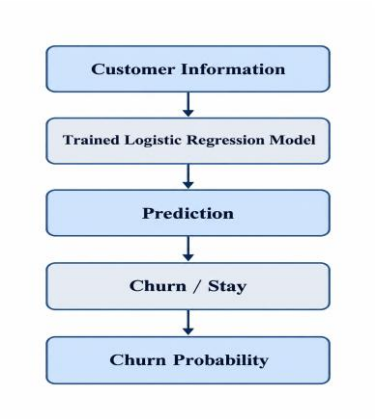


Figure 4: Inference Flowchart

The model can therefore be used after training without retraining every time a new customer is evaluated.

2.8 Graphical User Interface

A Gradio-based GUI was developed to make the system easier to use. Users can enter customer details using dropdowns, sliders, and numerical fields.

The interface produces an output such as:

Customer Likely To Churn

Churn Probability: 65.23%

This allows the ML model to be demonstrated as an application rather than only through Python code.

CHAPTER 3:EXPERIMENTS

3.1 Experimental Environment

The project was implemented using Python in Google Colab. Libraries such as Pandas, NumPy, Scikit-learn, Matplotlib, and Gradio were used for data processing, machine learning, visualization, and GUI development.

3.2 Dataset Preparation

The Telco Customer Churn dataset was loaded and preprocessed. The customerID column was removed, TotalCharges was converted to numeric form, and missing values were handled using median imputation. Categorical features were encoded using One-Hot Encoding, and numerical features were standardized. The dataset was divided into 80% training data and 20% testing data using a stratified train-test split.

3.3 Model Training

Three machine learning classification models were trained:

- Logistic Regression
- Decision Tree
- Random Forest

The dataset was divided into 80% training data and 20% testing data.

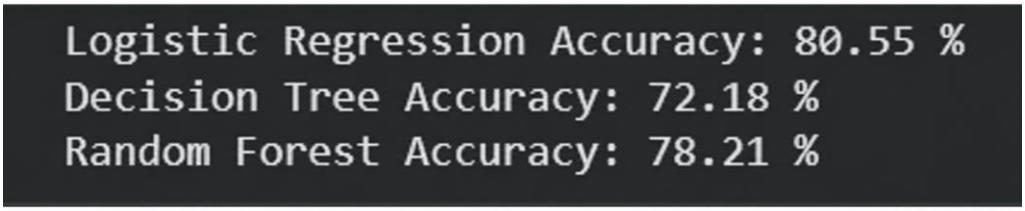


Figure 5: Model Training Output

3.4 Model Comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression	80.55	65.72	55.88	60.40
Decision Tree	72.18	47.76	51.34	49.48
Random Forest	78.21	61.13	49.20	54.52

The models were compared using accuracy, precision, recall, and F1-score. The final model was selected based on the highest F1-score to balance precision and recall. The selected model was then used for customer churn prediction and integrated into the Gradio interface.

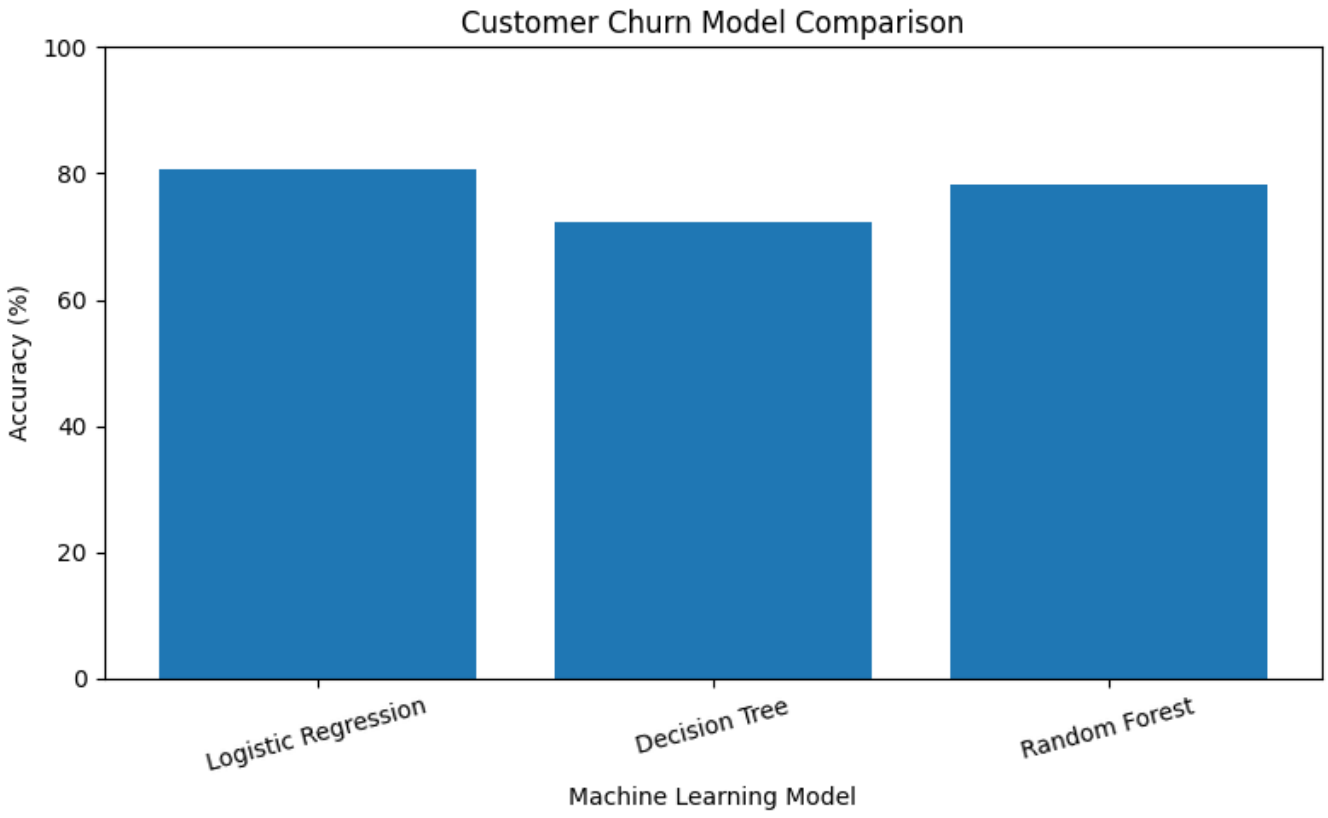


Figure 6: Model Accuracy Comparison

3.5 Inference Experiment

The trained Logistic Regression model was used to predict new customer records. The system provides the predicted Churn/Stay status along with the churn probability.

```
sample = X_test.iloc[[0]]

prediction = lr.predict(sample)
probability = lr.predict_proba(sample)

print("Prediction:", prediction[0])
print("Churn Probability:", probability[0][1]*100)
```

```
Prediction: 1
Churn Probability: 51.39417100800303
```

Figure 7: Customer Churn Prediction System Flowchart.

CHAPTER 4:RESULTS

4.1 Model Performance

The trained Logistic Regression, Decision Tree, and Random Forest models were evaluated using the test dataset. Their performance was measured using accuracy, precision, recall, F1-score, and confusion matrices.

4.2 Model Selection

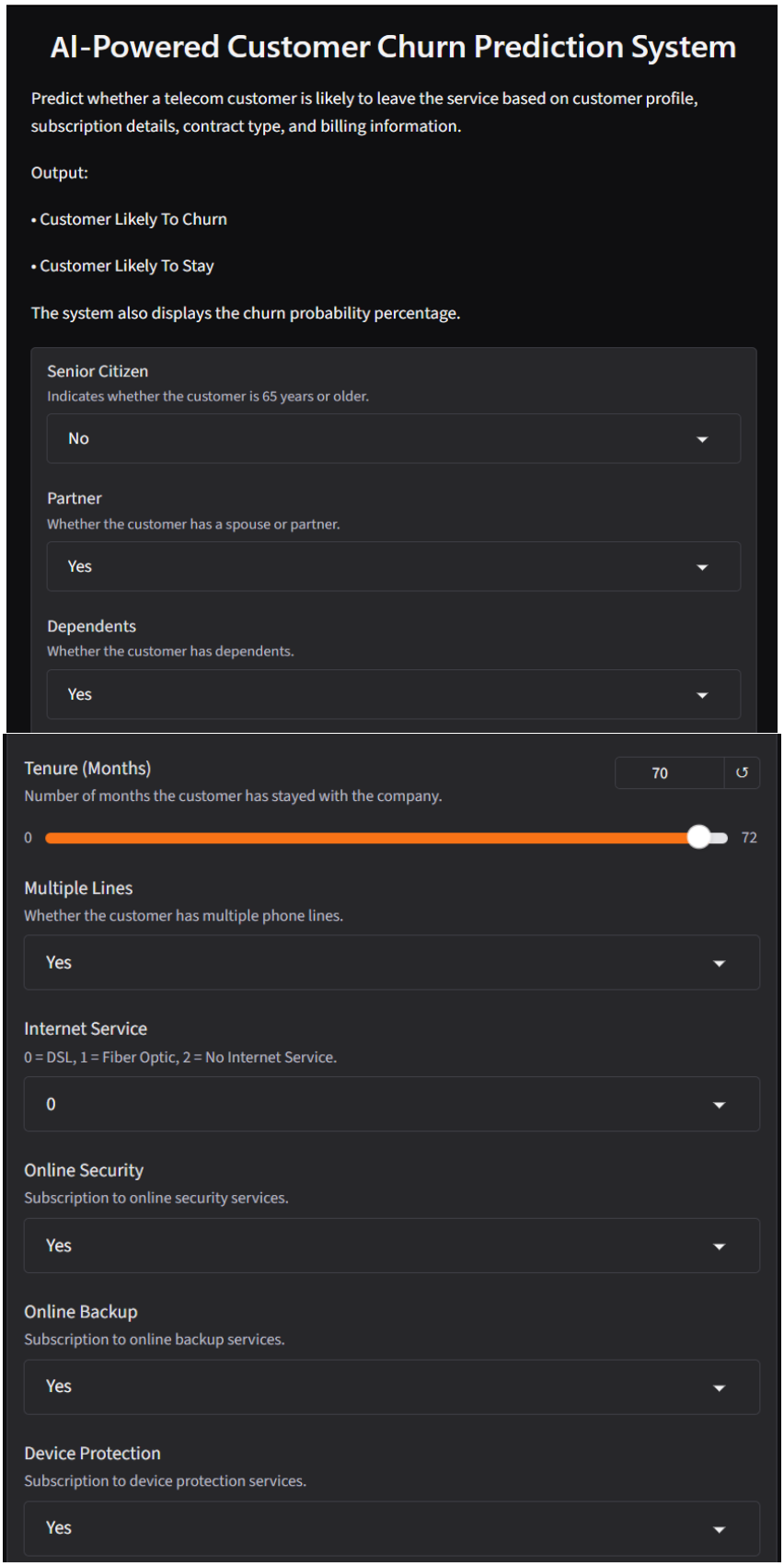
The models were compared based on their evaluation metrics, and the model with the highest F1-score was selected for customer churn prediction. The selected model was used to predict whether a customer is likely to churn or stay.

4.3 Inference Results

The selected model predicts whether a customer is likely to Churn or Stay and provides the corresponding churn probability.

4.4 GUI Results

A Gradio-based GUI was developed to allow users to enter customer details and obtain the churn prediction easily.


AI-Powered Customer Churn Prediction System
Predict whether a telecom customer is likely to leave the service based on customer profile, subscription details, contract type, and billing information.
Output:
• Customer Likely To Churn
• Customer Likely To Stay
The system also displays the churn probability percentage.
Senior Citizen
Indicates whether the customer is 65 years or older.
No
Partner
Whether the customer has a spouse or partner.
Yes
Dependents
Whether the customer has dependents.
Yes
Tenure (Months)
Number of months the customer has stayed with the company.
70
0
72
Multiple Lines
Whether the customer has multiple phone lines.
Yes
Internet Service
0 = DSL, 1 = Fiber Optic, 2 = No Internet Service.
0
Online Security
Subscription to online security services.
Yes
Online Backup
Subscription to online backup services.
Yes
Device Protection
Subscription to device protection services.
Yes

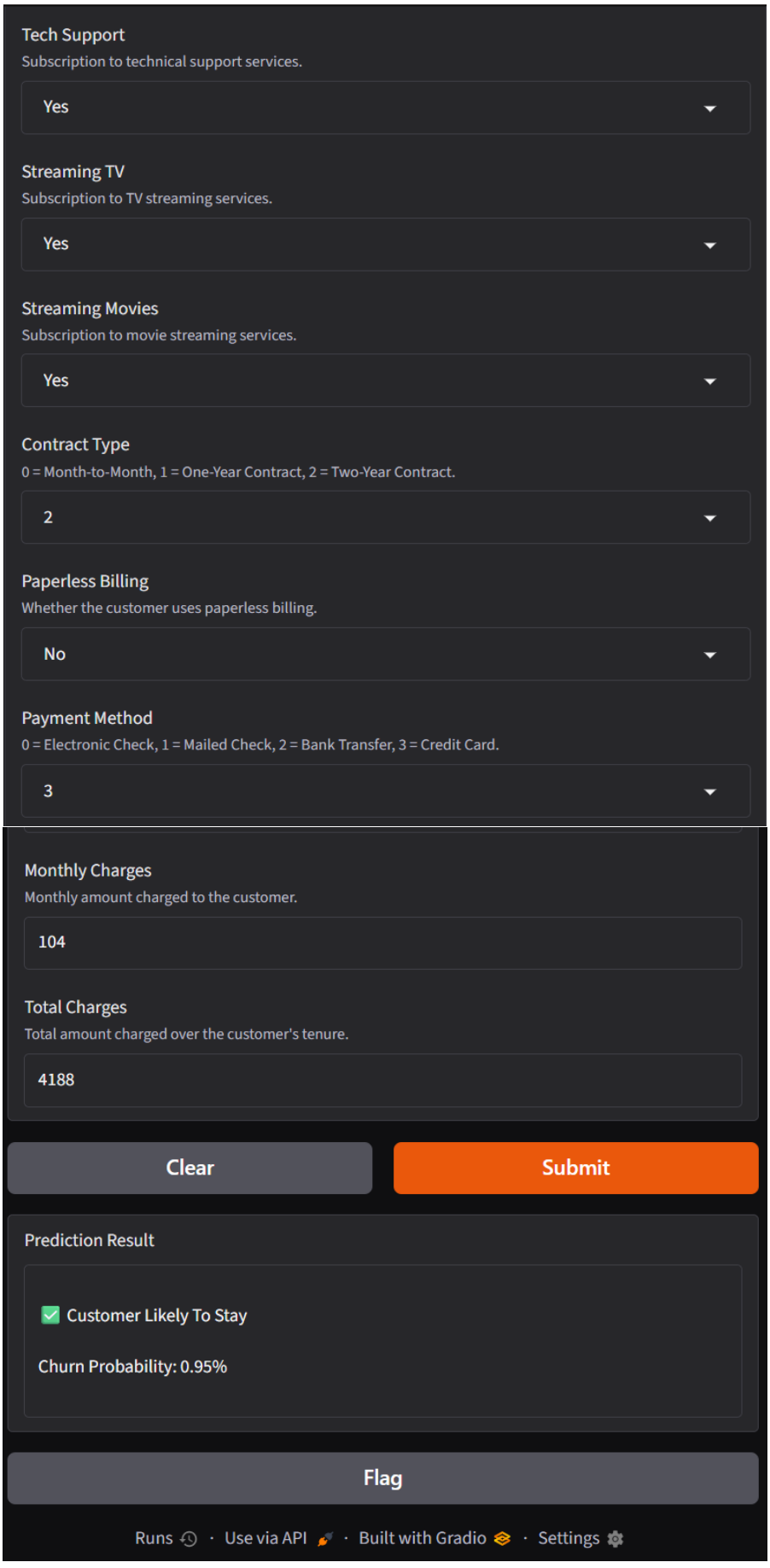


Figure 8: Customer Churn Prediction GUI

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

The project developed a machine learning-based customer churn prediction system using the Telco Customer Churn dataset. Logistic Regression, Decision Tree, and Random Forest were trained and evaluated using multiple performance metrics. The best-performing model was selected based on F1-score and integrated into a Gradio-based graphical user interface. The system predicts customer churn and provides a churn probability, helping demonstrate how machine learning can support customer retention decisions.

5.2 Future Work

- Use advanced machine learning algorithms to improve prediction accuracy.
- Apply hyperparameter tuning for better model performance.
- Add Explainable AI to understand the reasons behind churn predictions.
- Integrate real-time customer data for continuous prediction.
- Develop automated retention recommendations for high-risk customers.
- Improve the GUI with additional visualization and customer insights.

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USE REFERENCES SUCH AS:

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